

Calculation of Two-Dimensional Vibrational Potential Energy Surfaces Utilizing Prediagonalized Basis Sets and Van Vleck Perturbation Methods

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(Received: April 25, 1985)

This work reviews the development of the vibrational Hamiltonian from fundamental principles and elaborates on the theoretical calculation of kinetic and potential energy functions for two interacting vibrations. Detailed procedures for determining kinetic energy expansions and energy levels using prediagonalized basis functions are presented. The necessary matrix elements are tabulated. The utility of the Van Vleck transformation method for facilitating the calculation of two-dimensional potential energy surfaces is demonstrated.

Introduction

For several years a considerable amount of effort has been devoted to studying the conformational dynamics of small ring molecules.¹⁻⁷ A variety of different four- and five-membered and pseudo-four- and pseudo-five-membered ring molecules have had their potential energy functions determined for the ring-puckering vibration. This motion typically is capable of taking a molecule from one conformation to another, and its potential energy function is thus highly informative. The greatest amount of research has been done assuming that the ring-puckering vibration can be considered separately from the other vibrational modes (one-dimensional treatment) because of its low frequency.

An understanding of the true conformational dynamics of small ring molecules can only be ascertained by considering the influence of other vibrational modes on the ring-puckering vibration. Such interaction can be observed spectroscopically as a shifting of the ring-puckering frequencies in the excited state of another vibration. For example, we have recently reported the interaction of the ring-puckering vibration with the PH inversion vibration in 3-phospholene,^{8,9} with the ring-twisting vibration in cyclopentene,¹⁰ and with the SiH₂ in-phase rocking and ring deformation vibrations in 1,3-disilacyclobutane.¹¹ In analyzing the interaction of the PH inversion and ring-puckering vibrations, we used and developed a variety of theoretical methods in determining the two-dimensional potential energy surface for these vibrations. It is the intent of this paper to present a brief review of the development of the vibrational Hamiltonian from fundamental principles and to elaborate on the theoretical calculation of kinetic and potential energy functions for the interaction of two vibrations (expanding the treatment to the interaction of three, four, etc., vibrations would follow directly). In particular, the application of Van Vleck perturbation methods to these types of problems will be presented. Previous treatment of the theoretical background for studying the interaction of the ring puckering (or in general a large-amplitude vibration) with another vibration has not been given completely. We will attempt to do so here on the basis of methods we have used in analyzing two-dimensional problems.

Theoretical Development of Vibrational Hamiltonian

General Development of the Internal Motion Hamiltonian for Molecules. The development and selection of an appropriate Hamiltonian, which is a representation of the total energy of a system, is a formidable task. The Hamiltonian is given by

$$H = T + V \quad (1)$$

where T is the kinetic energy dependent on the mass and velocity of the system and V is the potential energy, which is a function of the displacement from equilibrium.

Wilson, Decius, and Cross¹² have shown (pp 273-5) that the

kinetic energy may be described by

$$2T = \dot{\vec{R}}^2 \sum_{\alpha=1}^n m_{\alpha} + \sum_{\alpha=1}^n m_{\alpha} (\vec{\omega} \times \vec{r}_{\alpha}) \cdot (\vec{\omega} \times \vec{r}_{\alpha}) + \sum_{\alpha=1}^n m_{\alpha} \vec{v}_{\alpha}^2 + 2\vec{\omega} \cdot \sum_{\alpha=1}^n m_{\alpha} (\vec{p}_{\alpha} \times \vec{v}_{\alpha}) \quad (2)$$

where m_{α} and $\vec{v}_{\alpha}$ are the mass and velocity of the α th atom, $\dot{\vec{R}}$ is the time derivative of a vector connecting the origin of the coordinate system to the center of mass, $\vec{\omega}$ is the angular velocity for the rotating coordinate system, $\vec{r}_{\alpha}$ is a vector from the center of mass to the α th atom of the molecule, and $\vec{p}_{\alpha}$ is a displacement vector reflecting the change in $\vec{r}_{\alpha}$ from its equilibrium representation. As an example, 2 of the 12 coordinate vectors for the 3-phospholene molecule, $\vec{r}_2$ and $\vec{r}_9$, are illustrated in Figure 1.

The four terms in eq 2 are respectively the translational kinetic energy, the rotational kinetic energy, the vibrational kinetic energy, and the interaction kinetic energy of the vibrations and rotations. The significant feature to note about eq 2 is that the translational kinetic energy is totally separable from that of the vibrations and rotations.

If one uses reduction formulas for vector triple products, neglects the translational kinetic energy, and introduces internal vibration coordinates, eq 2 becomes in matrix notation¹³

$$2T = (\omega^t, \dot{q}^t) \begin{pmatrix} \mathbf{I} & \mathbf{X} \\ \mathbf{X}^t & \mathbf{Y} \end{pmatrix} \begin{pmatrix} \omega \\ \dot{q} \end{pmatrix} \quad (3)$$

where ω is a column vector of the three angular velocity components and $\dot{q}$ is a $(3N - 6)$ -dimensional column vector of the time derivatives of the vibrational coordinates. The matrix $\mathbf{I}$ is the 3×3 instantaneous inertial tensor

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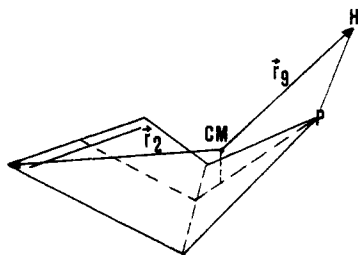


Figure 1. Equilibrium conformation of 3-phospholene showing the center of mass (CM) and two of the coordinate vectors.

$$\mathbf{I} = \begin{pmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{pmatrix} \quad (4)$$

where the elements are derived from the relationships

$$I_{kk} = \sum_{\alpha=1}^n m_{\alpha} (\bar{r}_{\alpha}^2 - r_{\alpha k}^2); \quad k = x, y, \text{ or } z \quad (5)$$

and

$$I_{kk'} = \sum_{\alpha=1}^n m_{\alpha} r_{\alpha k} r_{\alpha k'}; \quad k \neq k' \quad (6)$$

As an example, the $\mathbf{X}$ matrix is a $3 \times (3N - 6)$ matrix with the elements defined by

$$X_{ij} = \sum_{\alpha=1}^n m_{\alpha} [\bar{r}_{\alpha} \times (\partial \bar{r}_{\alpha} / \partial q_j)]_i \quad (7)$$

Finally, the $\mathbf{Y}$ matrix is a $(3N - 6) \times (3N - 6)$ matrix with elements derived from the expression

$$Y_{ij} = \sum_{\alpha=1}^n m_{\alpha} \left(\frac{\partial \bar{r}_{\alpha}}{\partial q_i} \right) \cdot \left(\frac{\partial \bar{r}_{\alpha}}{\partial q_j} \right) \quad (8)$$

The q_i 's and q_j 's in eq 7 and 8 are the vibrational coordinates.

Equation 3 may be transformed into momentum representation by simply inverting the middle matrix on the right-hand side of the equation.^{13,14}

$$2T = (\mathbf{p}^t, \mathbf{p}^t) \begin{pmatrix} \mathbf{I} & \mathbf{X} \\ \mathbf{X}^t & \mathbf{Y} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{p}^t \\ \mathbf{p}^t \end{pmatrix} \quad (9)$$

Here, $\mathbf{P}$ is a 3-dimensional column vector defined by

$$\mathbf{P} = (\partial T / \partial \omega) \quad (10)$$

and $\mathbf{p}$ is a $(3N - 6)$ -dimensional column vector of the momenta conjugate to q , the vibrational coordinate

$$\mathbf{p} = (\partial T / \partial \dot{q}) \quad (11)$$

In conventional notation, we may write eq 9 as

$$2T = (\mathbf{P}^t, \mathbf{p}^t) \mathbf{G} \begin{pmatrix} \mathbf{P} \\ \mathbf{p} \end{pmatrix} \quad (12)$$

where

$$\mathbf{G} = \begin{pmatrix} \mathbf{I} & \mathbf{X} \\ \mathbf{X}^t & \mathbf{Y} \end{pmatrix}^{-1} \quad (13)$$

Equation 13 is termed the rotational-vibrational $\mathbf{G}$ matrix, and $\mathbf{Y}$, the pure vibrational portion of this matrix, is invariant with the choice of the axis system.¹³

We now wish to transform the kinetic energy given in eq 12 to the quantum-mechanical operator form. The procedure for transforming to this form is given elsewhere^{15,16} and will not be considered in detail here. The quantum-mechanical form for the

vibrational-rotational kinetic energy is given by

$$2T = g^{-1/4} \sum_{i=1}^3 \sum_{j=1}^3 P_i g^{1/2} g_{ij} P_j g^{-1/4} + g^{-1/4} \sum_{i=1}^3 \sum_{j=4}^{3N-3} (P_i g^{1/2} g_{ij} P_j + P_j g^{1/2} g_{ji} P_i) g^{-1/4} + g^{-1/4} \sum_{k=4}^{3N-3} \sum_{l=4}^{3N-3} P_k g^{1/2} g_{kl} P_l g^{-1/4} \quad (14)$$

where g is the determinant of the $\mathbf{G}^{-1}$ matrix and the g_{ij} elements are the elements taken from eq 13 after all the computations have been performed. The first and third terms in eq 14 represent the pure rotational and vibrational kinetic energy, respectively. The second term arises from the interaction between vibrations and rotations, which leads to the Coriolis energy.

If we are interested only in vibrational transitions between the $J = 0$ rotational states (as is assumed), we need only consider the last term in eq 14 and we have

$$2T_{\text{VIB}} = g^{-1/4} \sum_{k=4}^{3N-3} \sum_{l=4}^{3N-3} P_k g^{1/2} g_{kl} P_l g^{-1/4} \quad (15)$$

This is equivalent to¹⁴

$$T_{\text{VIB}} = -\frac{\hbar^2}{2} \sum_{k=1}^{3N-6} \sum_{l=1}^{3N-6} \left[\frac{\partial}{\partial q_k} + \frac{1}{4} \left(\frac{\partial \ln g}{\partial q_k} \right) \right] g_{kl} \left[\frac{\partial}{\partial q_l} - \frac{1}{4} \left(\frac{\partial \ln g}{\partial q_l} \right) \right] \quad (16)$$

Upon rearrangement, this expression for the vibrational kinetic energy can be written as

$$T_{\text{VIB}} = -\frac{\hbar^2}{2} \sum_{k=1}^{3N-6} \sum_{l=1}^{3N-6} \frac{\partial}{\partial q_k} g_{kl} \frac{\partial}{\partial q_l} + V' \quad (17)$$

where V' is the "pseudopotential" given by

$$V' = \frac{1}{8} \hbar^2 \sum_{k=1}^{3N-6} \sum_{l=1}^{3N-6} \left[\frac{\partial}{\partial q_k} g_{kl} \left(\frac{\partial \ln g}{\partial q_l} \right) - \left(\frac{\partial \ln g}{\partial q_k} \right) g_{kl} \frac{\partial}{\partial q_l} \right] + \frac{1}{32} \hbar^2 \sum_{k=1}^{3N-6} \sum_{l=1}^{3N-6} \left(\frac{\partial \ln g}{\partial q_k} \right) g_{kl} \left(\frac{\partial \ln g}{\partial q_l} \right) \quad (18)$$

The Hamiltonian for the vibrations of a molecule in center-of-mass coordinates has the form

$$H_{\text{VIB}} = -\frac{\hbar^2}{2} \sum_{k=1}^{3N-6} \sum_{l=1}^{3N-6} \frac{\partial}{\partial q_k} g_{kl} \frac{\partial}{\partial q_l} + V' + V(q_1, q_2, \dots) \quad (19)$$

where $V(q_1, q_2, \dots)$ is the potential energy. The form of this potential energy function will be considered later.

Approximations Used in Studying Large-Amplitude Vibrations. A large-amplitude ring-puckering vibration can, in general, only be poorly described with a harmonic potential function. As a result, the standard Wilson FG method used for studying other molecular vibrations would prove inadequate.¹² Typically, the large-amplitude vibrations have low frequencies. In many cases each large-amplitude mode (such as the ring-puckering vibration) can, to a good first approximation, be considered to be independent of all other $3N - 7$ vibrational modes of the molecule (the high-low frequency separation approximation). However, spectroscopic evidence shows that this approximation is sometimes not entirely valid because interaction between the ring puckering and other vibrations can exist.^{8-11,17,18}

The assumption is also made that the pseudopotential terms given in eq 18 are small enough to be neglected. Calculations have been reported substantiating this assumption.¹⁹ We will not assume that the g_{kl} elements in eq 19 commute with the momentum operator (i.e., $[\partial / \partial q_k, g_{kl}] \neq 0$). However, most studies involving the ring-puckering vibration have made the

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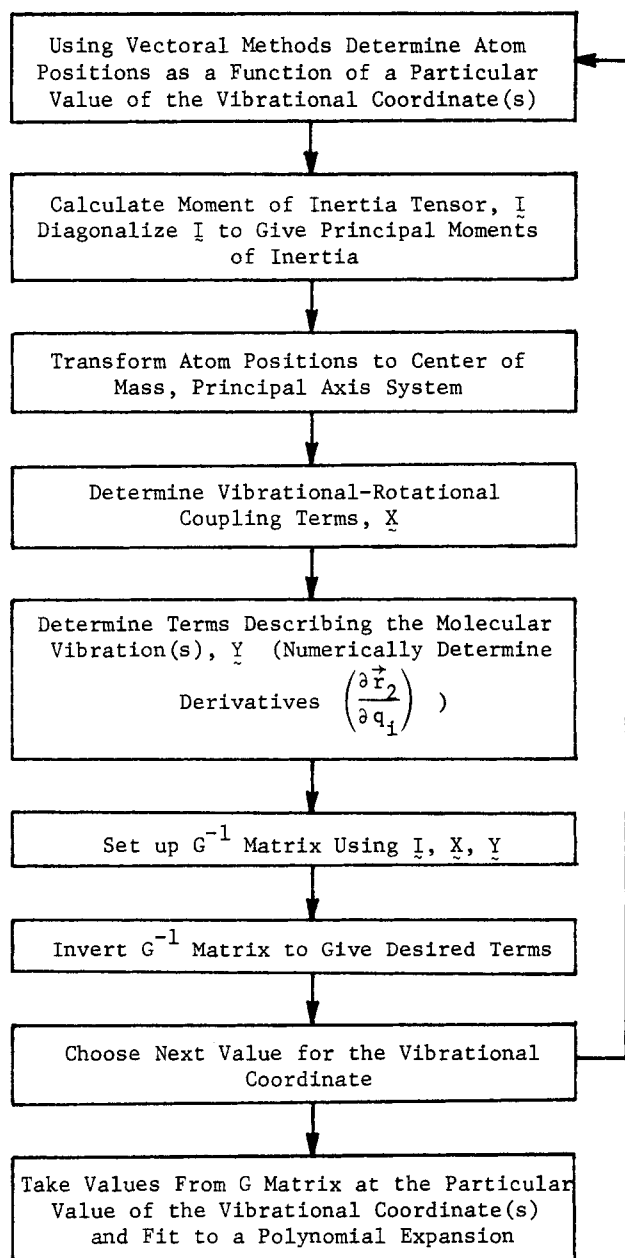


Figure 2. Outline of procedure of determining kinetic energy expansions.

assumption of a constant reduced mass.

Determination of Kinetic Energy Functions. Determination of kinetic energy functions or expansions for the vibrational Hamiltonian will be considered in two ways. First, the expansions will be determined by assuming that no coupling of a ring-puckering vibration occurs with other vibrations and, second, that coupling does exist.

In order to better understand kinetic energy calculations when considering a two-dimensional problem, we will first assume that the large-amplitude ring-puckering vibration is separable from the other $3N - 7$ vibrations (i.e., no coupling). By use of this assumption, the Hamiltonian has the simple form (neglecting pseudopotential terms)

$$H_{\text{VIB}} = -\frac{\hbar^2}{2} \frac{\partial}{\partial x} g_{44}(x) \frac{\partial}{\partial x} + V(x) \quad (20)$$

where x is the (curvilinear) ring-puckering coordinate, $g_{44}(x)$ is the kinetic energy term or expansion from the rotational-vibrational G matrix for the ring-puckering vibration, and $V(x)$ is the potential energy function.

The g_{44} element must be determined, and eq 13 may be used. For a one-dimensional calculation, the G matrix has the form

$$G = \begin{pmatrix} I_{xx} & -I_{xy} & -I_{xz} & X_{11} \\ -I_{xy} & I_{yy} & -I_{yz} & X_{12} \\ -I_{xz} & -I_{yz} & I_{zz} & X_{13} \\ X_{11} & X_{12} & X_{13} & Y_{11} \end{pmatrix}^{-1} \quad (21)$$

where Y_{11} is a vibrational term and the other elements in the equation may be determined from eq 5-8. The derivatives $\partial \vec{r}_i / \partial q_l$ are usually determined by using numerical methods. A complete explanation of the calculation of the matrix for small ring molecules has been given elsewhere.^{20,21}

After the matrix inversion is performed in eq 21, the G matrix may be written as

$$G = \begin{pmatrix} g_{11} & g_{12} & g_{13} & g_{14} \\ g_{12} & g_{22} & g_{23} & g_{24} \\ g_{13} & g_{23} & g_{33} & g_{34} \\ g_{14} & g_{24} & g_{34} & g_{44} \end{pmatrix} \quad (22)$$

where the g_{44} element is the pure vibrational term needed for determining the $g_{44}(x)$ expression in eq 20.

The ring-puckering vibration is a large-amplitude vibration resulting in considerable structural changes. Therefore, the g_{44} term cannot be considered to be independent of the puckering vibrational coordinate.^{22,23} The coordinate dependence of g_{44} can be taken into consideration by forming a polynomial expansion of g_{44} as function of the vibrational coordinate (i.e. $g_{44}(x)$). The general form of this expansion is

$$g_{44}(x) = \sum_{l=1}^m g_{44}^{(l)} x^l = 1/\mu \quad (23)$$

where the $g_{44}^{(l)}$ are the expansion coefficients and μ is the reduced mass. Usually the expansion is taken to 4th power for asymmetric vibrations and 6th power for symmetric vibrations (i.e. $m = 4$ or 6).

The coefficients in eq 23 are found by forming the G matrix in eq 22 at various values of the vibrational coordinate. That is, a value of the puckering coordinate is chosen which determines a structure for the molecule and then the G matrix is determined. This procedure is typically repeated 20-25 times for several values for the puckering coordinate. The Newton-Raphson procedure may then be used to determine the expansion coefficients.²⁴ The general procedure is outlined in Figure 2.

In an analysis of the ring-puckering vibration coupled with a second vibration, the vibrational Hamiltonian has the form (neglecting pseudopotential terms)

$$H_{\text{VIB}} = -\frac{\hbar^2}{2} \left[\frac{\partial}{\partial x} g_{44}(x,y) \frac{\partial}{\partial x} + \frac{\partial}{\partial x} g_{45}(x,y) \frac{\partial}{\partial y} + \frac{\partial}{\partial y} g_{45}(x,y) \frac{\partial}{\partial x} + \frac{\partial}{\partial y} g_{55}(x,y) \frac{\partial}{\partial y} \right] + V(x,y) \quad (24)$$

where $g_{44}(x,y)$, $g_{45}(x,y)$, and/or $g_{55}(x,y)$ may be functions of the x and/or y coordinates (or may not be a function of either coordinate).

In order to determine the form of $g_{44}(x,y)$, $g_{45}(x,y)$, and $g_{55}(x,y)$, several items must be considered. First, are the G matrix terms for the two vibrations coordinate dependent? If so, which coordinates should be used for the function? Also, are the two vibrations "significantly" coupled kinematically (i.e., is $g_{45} \neq 0$)? Finally, what are the symmetry conditions governing the form of the functions?

The G matrix for two vibrations has the form

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$$G = \begin{pmatrix} I_{xx} & -I_{xy} & -I_{xz} & X_{11} & X_{21} \\ -I_{xy} & I_{yy} & -I_{yz} & X_{12} & X_{22} \\ -I_{xz} & -I_{yz} & I_{zz} & X_{13} & X_{23} \\ X_{11} & X_{12} & X_{13} & Y_{11} & Y_{12} \\ X_{21} & X_{22} & X_{23} & Y_{12} & Y_{22} \end{pmatrix}^{-1} \quad (25)$$

After matrix inversion, the matrix becomes

$$G = \begin{pmatrix} g_{11} & g_{12} & g_{13} & g_{14} & g_{15} \\ g_{12} & g_{22} & g_{23} & g_{24} & g_{25} \\ g_{13} & g_{23} & g_{33} & g_{34} & g_{35} \\ g_{14} & g_{24} & g_{34} & g_{44} & g_{45} \\ g_{15} & g_{25} & g_{35} & g_{45} & g_{55} \end{pmatrix} \quad (26)$$

where g_{44} , g_{45} , and g_{55} are the pure vibrational terms which are necessary for determining the kinetic energy functions in eq 24. The g_{44} term for the ring-puckering vibration as determined from eq 25 will not, in general, be the same as determined by eq 21 if the two vibrations are of the same symmetry. This has been shown and discussed previously.^{12,21}

The most general expression for $g_{44}(x,y)$, $g_{45}(x,y)$, and $g_{55}(x,y)$ as a function of the two vibrational coordinates is given by

$$g_{ij}(x,y) = \sum_{l=0}^n \sum_{k=0}^m g_{ij}^{(l,k)} x^l y^k \quad (27)$$

where $i = 4, 5$ and $j = 4, 5$. The coefficients, $g_{ij}^{(l,k)}$, are determined in much the same way as the coefficients in eq 23. However, in this case, a grid of points at various combinations of values for x and y must be determined. Then, a least-squares procedure involving two independent and one dependent variable can be used to determine the coefficients.

The least-squares technique involves minimizing the squared difference between "observed" and calculated values. In this case, we select values of x and y and determine "observed" values for g_{44} , g_{45} , and g_{55} . Separate expansions using the least-squares method are determined for $g_{44}(x,y)$, $g_{45}(x,y)$, and $g_{55}(x,y)$. The least-squares equation is given by

$$\eta = \sum_{j=1}^N (f - \bar{f})^2 \quad (28)$$

where N is the number of data points, f is the dependent variable (in our case g_{44} , g_{45} , or g_{55}), and

$$\bar{f} = \sum_{k=1}^m g_k x^{l_k} y^{n_k} \quad (29)$$

where g_k are the polynomial expansion coefficients as given in eq 27, m is the number of coefficients, and l_k and n_k indicate the exponents for the x and y coordinate, respectively, for the k th coefficient. If we minimize η with respect to the coefficients, we get

$$d\eta = 0 = \sum_i^m (\partial\eta/\partial g_i) dg_i \quad (30)$$

We can find $(\partial\eta/\partial g_i)$ from eq 28:

$$\partial\eta/\partial g_i = 2 \sum_{j=1}^N [f_j - \bar{f}_j] x_j^{l_i} y_j^{n_i} = 0 \quad (31)$$

Substitution of $\bar{f}$ as defined in eq 29 results in

$$\sum_{j=1}^N f_j x_j^{l_i} y_j^{n_i} = \sum_{k=1}^m g_k \sum_{j=1}^N x_j^{(l_i+l_k)} y_j^{(n_i+n_k)} \quad (32)$$

The matrix form of the equation is

$$\Gamma = Z^{-1}F \quad (33)$$

The Z^{-1} matrix results from the second summation term on the right-hand side of eq 32, F is the matrix of the terms on the left-hand side, and Γ is the matrix of the coefficients that satisfies eq 27.

Potential Energy Functions. The potential energy function

describing a large-amplitude vibration, which is the key to understanding the molecule's conformational energies and dynamics, is determined by the nature and symmetry of the vibration. A potential energy function may be classified as periodic (e.g., internal rotation) or nonperiodic (e.g., ring-puckering, ammonia-like inversion, or small-amplitude vibrations). A periodic potential energy function may be written as a Fourier series

$$V(\phi) = \sum_{n=1}^k \frac{V_n}{2} (1 - \cos n\phi) \quad (34)$$

where ϕ is the angle of internal rotation and the V_n 's are the force constants.²⁵ Vibrations of this type will not be considered here. For treatments of periodic functions the reader is referred to ref 25 and 26 and the references therein.

The ring-puckering vibration (a nonperiodic vibration) can be described by a function containing the appropriate terms selected from a Taylor series expansion. For one independent variable x the Taylor series expansion for the potential energy is

$$V(x) = V_0 + \left(\frac{\partial V}{\partial x}\right)_{x=0} x + \frac{1}{2!} \left(\frac{\partial^2 V}{\partial x^2}\right)_{x=0} x^2 + \frac{1}{3!} \left(\frac{\partial^3 V}{\partial x^3}\right)_{x=0} x^3 + \dots \quad (35)$$

The V_0 term is a constant term which only affects the absolute value of the potential energy. The first-derivative term may be neglected if the minimum (or maximum) of the potential energy is chosen at the zero value of the vibrational coordinate. The potential function for a ring-puckering vibration which is symmetric about $x = 0$ (e.g. for trimethylene oxide and cyclopentene) can thus be written as

$$V(x) = ax^4 + bx^2 \quad (36)$$

where a is the force constant which is primarily related to ring strain forces and b is the force constant primarily related to the torsional forces.²⁷⁻³² For molecules with an asymmetric ring-puckering vibration (e.g., chlorocyclobutane, 2,5-dihydropyrrole, and 3-phospholene), the potential function has the form

$$V(x) = ax^4 + bx^2 + cx^3 \quad (37)$$

where c is also related to the torsional forces and is a measure of the asymmetry of the vibrational motion. Terms linear in x are normally excluded since the potential minimum ($V = 0$) can be selected to occur at $x = 0$.

In the case of multidimensional potential energy surfaces the form of the potential is not always easily determined. The functions are usually less well-defined than in one-dimensional treatments of the ring-puckering vibration. The complete two-dimensional potential energy function in terms of the vibrational coordinates x and y is given by

$$V(x,y) = \sum_{i=1}^m K_x^{(i)} x^i + \sum_{j=1}^n K_y^{(j)} y^j + \sum_{i'=1}^{n'} \sum_{j'=1}^{m'} K_{xy}^{(i'j')} x^{i'} y^{j'} \quad (38)$$

The K_x , K_y , and K_{xy} 's are force constants which may or may not be explicitly related to certain forces or interactions of vibrations in the ring as a , b , and c are in eq 37. This expression is only

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valid for vibrations which can be described by nonperiodic vibrational coordinates.

From the general expression in eq 38 for the potential energy surface, the appropriate terms are retained or dropped based on the symmetry conditions required for the interaction of the two vibrations and knowledge about the nature of the vibrations (e.g., harmonic or anharmonic). For example, the potential function used to describe the ring-puckering (x) and PH inversion (y) of 3 phospholene⁹ has the form

$$V(x,y) = a_1x^4 + b_1x^2 + a_2y^4 + b_2y^2 + a_{13}xy^3 + a_{22}x^2y^2 + a_{31}x^3y \quad (39)$$

where the potential constants are readily related to the K_x , K_y , and K_{xy} of eq 38.

The force constants in the one- or multidimensional potential functions involving the ring-puckering vibration are the unknown parameters to be determined from the experimental data. Initial guesses of the force constants are made, and as described later, a least-squares technique is used for determining the best set of force constants for the potential energy function of the ring-puckering vibration.

Calculation of Vibrational Energy Levels. Once kinetic energy functions have been determined and an initial set of force constants has been chosen, the vibrational energy levels for the ring-puckering vibration may be calculated. The calculation of the energy levels is necessary in order that a comparison can be made with the experimentally determined levels. The minimization of the difference between observed and calculated frequencies allows for the accurate determination of the potential energy surface. The energy levels are determined by using the variation method and matrix diagonalization techniques (e.g. the Givens-Householder method³³). Improved matrix diagonalization techniques have been developed in recent years.

The following discussion of the calculations of energy levels is limited to a two-dimensional vibrational problem. The procedure for performing one-dimensional calculations can easily be deduced from the discussion of the two-dimensional case. For convenience, calculations are usually done in cm^{-1} units. In order for the energy levels to be calculated in these units, the Hamiltonian must be transformed from dimensioned to dimensionless coordinates (this procedure is also followed if other energy units are to be used). The vibrational Hamiltonian in center-of-mass coordinates for two vibrations can be expanded from eq 19 (neglecting pseudo-potential terms) to give

$$H_{\text{vib}} = -\frac{1}{2} \hbar^2 \left[\frac{\partial}{\partial x} g_{44}(x,y) \frac{\partial}{\partial x} + \frac{\partial}{\partial x} g_{45}(x,y) \frac{\partial}{\partial y} + \frac{\partial}{\partial y} g_{45}(x,y) \frac{\partial}{\partial x} + \frac{\partial}{\partial y} g_{55}(x,y) \frac{\partial}{\partial y} \right] + \sum_r \sum_s K^{(r,s)} x^r y^s \quad (40)$$

where the last term is an alternate way of writing the potential energy function given in eq 38. The Hamiltonian is transformed to dimensionless coordinates by making the substitutions

$$x = Q_x/\alpha_x \quad \text{and} \quad y = Q_y/\alpha_y \quad (41)$$

where

$$\alpha_x = 2\pi \left(\frac{\nu_x}{h g_{44}^{(0,0)}} \right)^{1/2} \quad \text{and} \quad \alpha_y = 2\pi \left(\frac{\nu_y}{h g_{55}^{(0,0)}} \right)^{1/2} \quad (42)$$

ν_x and ν_y are the harmonic frequencies, h is Planck's constant, and $g_{44}^{(0,0)}$ and $g_{55}^{(0,0)}$ are the leading coefficients from the kinetic energy expansions for the x and y coordinates, respectively, as given in eq 27. The harmonic frequencies should be selected judiciously in order to minimize the basis set necessary for the calculations. This can be done semiquantitatively by selecting an energy value (e.g. 1000 cm^{-1}) at which the ring-puckering potential and the harmonic oscillator potential are to be coincident

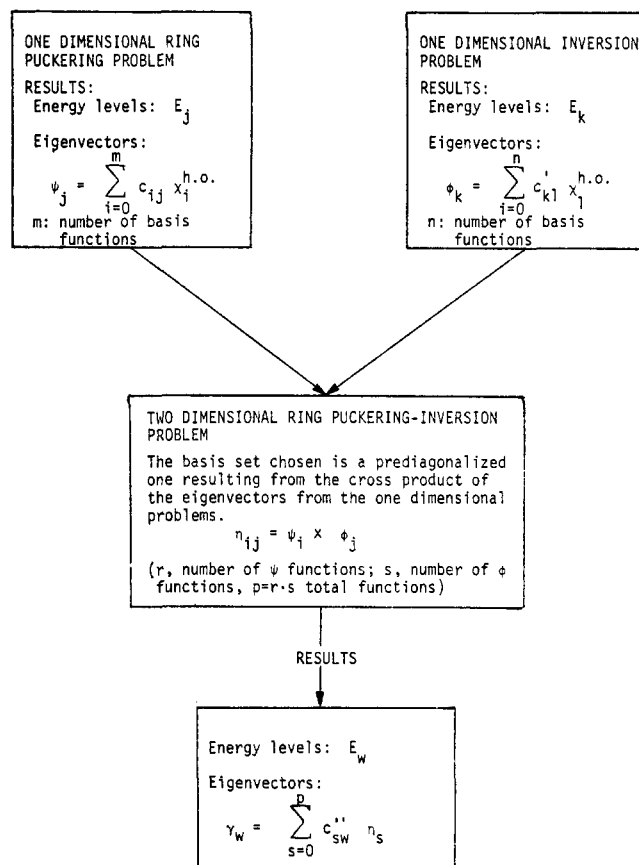


Figure 3. Outline of procedure for determining "prediagonalized" basis functions and calculating energy levels for a two-dimensional vibrational problem.

(this sets ν_x). The second harmonic frequency (ν_y) to be selected may often be taken as the observed vibrational frequency if the potential function for this coordinate has a single minimum.

The general expansions for $g_{44}(x,y)$, $g_{45}(x,y)$, and $g_{55}(x,y)$, can be substituted into eq 40. Use is then made of the commutation relationships

$$\left[\frac{\partial}{\partial x}, x^k \right] = kx^{k-1} \quad \text{and} \quad \left[\frac{\partial}{\partial y}, y^l \right] = ly^{l-1} \quad (43)$$

in order to arrive at the desired Hamiltonian. The transformed Hamiltonian that is used to compute the energy levels (in wavenumbers) is given by

$$H_{2D}(Q_x, Q_y) = \sum_{i=0}^u \sum_{j=0}^v \left\{ 0.5 \frac{\tilde{\nu}_x g_{44}^{(i,j)}}{g_{44}^{(0,0)}} \times \left[\left(\frac{g_{44}^{(0,0)}}{\tilde{\nu}_x} \right)^i \left(\frac{g_{55}^{(0,0)}}{\tilde{\nu}_y} \right)^j \right]^{1/2} b^{(i+j)} \right\} P_{Q_x} Q_x^i Q_y^j P_{Q_x} + \sum_{k=0}^{u'} \sum_{l=0}^{v'} \left\{ 0.5 g_{45}^{(k,l)} \left(\frac{g_{44}^{(0,0)}}{\tilde{\nu}_x} \right)^{(k-1)/2} \left(\frac{g_{55}^{(0,0)}}{\tilde{\nu}_y} \right)^{(l-1)/2} b^{(k+l)} \right\} \times \{ P_{Q_x} Q_x^k [P_{Q_y} Q_y^l + i_{\text{imag}} l Q_y^{l-1}] + P_{Q_y} Q_y^l [P_{Q_x} Q_x^k + i_{\text{imag}} k Q_x^{k-1}] \} + \sum_{m=0}^{u''} \sum_{n=0}^{v''} \left\{ 0.5 \frac{\tilde{\nu}_y g_{55}^{(m,n)}}{g_{55}^{(0,0)}} \left[\left(\frac{g_{44}^{(0,0)}}{\tilde{\nu}_x} \right)^m \left(\frac{g_{55}^{(0,0)}}{\tilde{\nu}_y} \right)^n \right]^{1/2} b^{(m+n)} \right\} \times P_{Q_y} Q_y^m Q_x^n P_{Q_y} + \sum_{r=0}^{u'''} \sum_{s=0}^{v'''} \left\{ K^{(r,s)} b^{(r+s)} \left(\frac{g_{44}^{(0,0)}}{\tilde{\nu}_x} \right)^{r/2} \left(\frac{g_{55}^{(0,0)}}{\tilde{\nu}_y} \right)^{s/2} \right\} Q_x^r Q_y^s \quad (44)$$

(33) J. M. Ortega in "Mathematical Methods for Digital Computers", Vol. 2, A. Ralston and T. H. Wolf, Eds., Wiley, New York, 1967, pp 94-115.

where i_{imag} is the imaginary number i and $b = (1/2\pi)(hN/c)^{1/2} = 5.80542 \text{ cm}^{-1}$.

Once the appropriate transformation of the Hamiltonian has been performed, the choice of basis functions must be made. Harmonic oscillator functions are often adequate for both one- and two-dimensional problems (see for example ref 10 and 27). For a one-dimensional problem the resulting wave functions have the form

$$\Psi_i(x) = \sum_{j=0}^n c_{ij} \phi_j^{\text{HO}} \quad (45)$$

where ϕ_k^{HO} are harmonic oscillator functions and the c_{ij} 's are the coefficients which are the variational parameters and constitute the eigenvectors for the problem. The harmonic oscillator wave functions are given by¹²

$$\Psi_v(x) = \alpha^{1/2} \pi^{-1/4} (2^v v!)^{-1/2} e^{-\alpha^2 x^2/2} H_v(\alpha x) \quad (46)$$

where $H_v(\alpha x)$ are the Hermite polynomials

$$H_v(\alpha x) = (-1)^v e^{\alpha^2 x^2} [d^v e^{-\alpha^2 x^2} / d(\alpha x)^v] \quad (47)$$

The harmonic frequencies, ν_x and ν_y , in eq 42, must be estimated, as described above, so that the most effective basis set can be obtained. For a two-dimensional analysis a harmonic oscillator basis set for both the x and y coordinate may be used. However, if the particular problem is poorly served by the use of harmonic oscillator basis sets for both coordinates, basis functions of the form given in eq 45 may be used for one or both of the coordinates (so-called "prediagonalized" functions). This may be the case if one or both of the vibrations being considered are anharmonic and expansions of harmonic oscillator functions can be calculated which better approximate the true wave functions for the system. Also, if one of the vibrations is asymmetric, the Hamiltonian matrix cannot be symmetry factored. The prediagonalized basis set provides a set of functions which are closer to the "true" functions. When these are used as a component of the basis set for a two-dimensional calculation, a significant reduction in the size of the Hamiltonian matrix can be achieved. Figure 3 shows the general procedure for determining and using the "prediagonalized" functions as a basis set. It should be noted that the cross product of two one-dimensional basis sets comprises the two-dimensional basis.

Once the basis set has been chosen in some form of a harmonic oscillator function, the Hamiltonian matrix from eq 44 may be determined by using the matrix elements for the various operators explicitly given in Tables I–III.

The matrix elements in these tables are determined in terms of the dimensionless coordinate $Q = \alpha^{1/2} x$, where α has been defined in eq 42 and from the following relationships:

$$(P)_{n,n+1} = -i[(n+1)/2]^{1/2} \quad (48)$$

$$(P)_{n,n-1} = i(n/2)^{1/2} \quad (49)$$

$$(Q)_{n,n+1} = [(n+1)/2]^{1/2} \quad (50)$$

$$(Q)_{n,n-1} = (n/2)^{1/2} \quad (51)$$

It should be noted that the following additional relationships are needed for determining the absolute value of the matrix elements for harmonic oscillator wave functions as given in Tables I–III:

$$A(r) = \left[\prod_{i=0}^r (n-i) \right]^{1/2} \quad (52)$$

$$B(t) = \left[\prod_{i=1}^t (n+i) \right]^{1/2} \quad (53)$$

The Hamiltonian matrix can now be diagonalized to give the energy levels and eigenvectors for the given kinetic energy expansion(s) and force constant(s).

Least-Squares Adjustment of Force Constants. In order to determine a set of force constants which predicts the observed frequencies, an iterative procedure for adjusting the force constants

is used.^{34,35} The procedure involves determining how the frequencies vary as a function of the potential force constants. We may write the dependence of a particular frequency, ϕ_i , as a function of n different force constants in terms of a Taylor series expansion

$$(\Delta\phi_i) \cong \Delta f_1 \frac{\partial \phi_i}{\partial f_1} + f_2 \frac{\partial \phi_i}{\partial f_2} + \dots + f_n \frac{\partial \phi_i}{\partial f_n} \quad (54)$$

However, for m frequencies eq 54 may be written in matrix notation as

$$\begin{pmatrix} \Delta\phi_1 \\ \Delta\phi_2 \\ \vdots \\ \Delta\phi_m \end{pmatrix} = \begin{pmatrix} \frac{\partial \phi_1}{\partial f_1} & \frac{\partial \phi_1}{\partial f_2} & \dots & \frac{\partial \phi_1}{\partial f_n} \\ \frac{\partial \phi_2}{\partial f_1} & \frac{\partial \phi_2}{\partial f_2} & \dots & \frac{\partial \phi_2}{\partial f_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \phi_m}{\partial f_1} & \frac{\partial \phi_m}{\partial f_2} & \dots & \frac{\partial \phi_m}{\partial f_n} \end{pmatrix} \begin{pmatrix} \Delta f_1 \\ \Delta f_2 \\ \vdots \\ \Delta f_n \end{pmatrix} \quad (55)$$

where the matrix containing the $\partial \phi_m / \partial f_n$ terms is known as the Jacobian matrix **J** and $\Delta\phi_i$ is the difference between observed and calculated frequencies. Equation 55 can be written in a simpler form:

$$\Delta\phi = \mathbf{J} \Delta\mathbf{F} \quad (56)$$

This method allows each frequency to be weighted, and the weighting factor may be based upon how well a particular frequency is known in relation to the other frequencies and/or the faith one has in the assignment of a frequency. The weighted matrix is represented by **W**. Multiplying eq 56 by **J^TW** and rearranging, we arrive at the commonly used equation for adjustment of the force constants

$$\Delta\mathbf{F} = (\mathbf{J}^T \mathbf{W} \mathbf{J})^{-1} \mathbf{J}^T \mathbf{W} \Delta\phi \quad (57)$$

where **W** is the weight matrix which allows each frequency to be considered equally or unequally with the other frequencies in the least-squares method.

In order to apply this technique, the Jacobian matrix must be determined. This in fact is the most difficult task in using the least-squares method. The Jacobian elements are a function of the frequencies which are in turn directly related to the eigenvalues of the system

$$J_{ij} = \left(\frac{\partial \phi_i}{\partial f_j} \right) = \left(\frac{\partial \lambda_k}{\partial f_j} \right) - \left(\frac{\partial \lambda_l}{\partial f_j} \right) \quad (58)$$

The eigenvalue λ_k of the system is found from

$$\lambda_k = \mathbf{L}_k^T \mathbf{H} \mathbf{L}_k \quad (59)$$

where $\mathbf{L}_k$ is the eigenvector corresponding to the k th eigenvalue and **H** is the Hamiltonian matrix. The Jacobian element becomes, after substitution of eq 59 for k and l eigenvalues into eq 58

$$J_{ij} = \left(\frac{\partial \phi_i}{\partial f_j} \right) = \mathbf{L}_k^T \left(\frac{\partial \mathbf{H}}{\partial f_j} \right) \mathbf{L}_k - \mathbf{L}_l^T \left(\frac{\partial \mathbf{H}}{\partial f_j} \right) \mathbf{L}_l \quad (60)$$

where $(\partial \mathbf{H} / \partial f_j)$ is an expectation value matrix over the coordinate corresponding to the f_j force constant. For example, if f_j is the force constant a in eq 37, then the matrix consists of expectation values for x^4 . Therefore, once the eigenvectors of the system are known and the observed and calculated frequencies are known, the least-squares method may be used to find a set of force constants that describe the conformational dynamics of the ring molecule.

(34) P. Gans, "Vibrating Molecules", Chapman and Hall, London, 1971.

(35) J. Inagaki, I. Harada, and T. Shimonouchi, *J. Mol. Spectrosc.*, **46**, 381 (1973).

TABLE I: Harmonic Oscillator Matrix Elements for PQ^i Operators^a

$\hat{O}$	$n+7$	$n+6$	$n+5$	$n+4$	$n+3$	$n+2$	$n+1$	n
P	0	0	0	0	0	0	$(-i/2^{1/2})B(1)$	0
PQ	0	0	0	0	0	$(-i/2)B(2)$	0	$i/2$
PQ^2	0	0	0	0	$[-i/2(2^{1/2})]B(3)$	0	$[-i/2(2^{1/2})](n-1)B(1)$	0
PQ^3	0	0	0	$(-i/4)B(4)$	0	$(-i/2)B(2)$	0	$(3i/4)(2n+1)$
PQ^4	0	0	$[-i/4(2^{1/2})]B(5)$	0	$[-i/4(2^{1/2})](3n+2)B(3)$	0	$[-i/4(2^{1/2})](2n^2-8n-9)B(1)$	0
PQ^5	0	$(-i/8)B(6)$	0	$(-i/8)(4n+5)B(4)$	0	$(-5i/8)(n^2-n-3)B(2)$	0	$(15i/8)(2n^2+2n+1)$
PQ^6	$[-i/8(2^{1/2})]B(7)$	0	$[-i/8(2^{1/2})](5n+9)B(5)$	0	$[-3i/8(2^{1/2})](3n^2+2n-5)B(3)$	0	$[-5i/8(2^{1/2})](n^3-9n^2-19n-15)B(1)$	0

 m

$\hat{O}$	$n-1$	$n-2$	$n-3$	$n-4$	$n-5$	$n-6$	$n-7$
P	$(i/2^{1/2})A(0)$	0	0	0	0	0	0
PQ	0	$(i/2)A(1)$	0	0	0	0	0
PQ^2	$[i/2(2^{1/2})](n+2)A(0)$	0	$[i/2(2^{1/2})]A(2)$	0	0	0	0
PQ^3	0	$(i/2)(n+1)A(1)$	0	$(i/4)A(3)$	0	0	0
PQ^4	$[i/4(2^{1/2})](2n^2+12n+1)A(0)$	0	$[i/4(2^{1/2})](3n+1)A(2)$	0	$[i/4(2^{1/2})]A(4)$	0	0
PQ^5	0	$(5i/8)(n^2+3n-1)A(1)$	0	$(i/8)(4n-1)A(3)$	0	$(i/8)A(5)$	0
PQ^6	$[5i/8(2^{1/2})](n^3+12n^2+2n+6)A(0)$	0	$[3i/8(2^{1/2})](3n^2+4n-4)A(2)$	0	$[i/8(2^{1/2})](5n-4)A(4)$	0	$[i/8(2^{1/2})]A(6)$

^a Matrix elements are represented by $\langle n|\hat{O}|m\rangle$; $\hat{O}$ is the operator PQ^i ; $i = (-1)^{1/2}$ and the $A(r)$ and $B(r)$ terms are defined in eq 52 and 53.TABLE II: Harmonic Oscillator Matrix Elements for PQ^iP Operators^a

$\hat{O}$	$n+8$	$n+7$	$n+6$	$n+5$	$n+4$	$n+3$	$n+2$	$n+1$	n
P^2	0	0	0	0	0	0	$(-1/2)B(2)$	0	$(2n+1)/2$
PQP	0	0	0	0	0	$[-1/2(2^{1/2})]B(3)$	0	$[1/2(2^{1/2})](B(1))^3$	0
PQ^2P	0	0	0	0	$(-1/4)B(4)$	0	0	$[1/4(2^{1/2})](2n^2+4n+9)B(1)$	$(1/4)(2n^2+2n+3)$
PQ^3P	0	0	0	$[-1/4(2^{1/2})]B(5)$	0	$[1/4(2^{1/2})](n+2)B(3)$	0	0	0
PQ^4P	0	$(-1/8)B(6)$	0	$(-1/8)(2n+5)B(4)$	0	$(1/8)(n^2+3n+15)B(2)$	0	$(1/8)(4n^3+6n^2+15)$	$(1/8)(4n^3+6n^2+32n)$
PQ^5P	0	$[-1/8(2^{1/2})]B(7)$	0	$[-3/8(2^{1/2})](n+3)B(5)$	0	$[-1/8(2^{1/2})](n^2+4n-15)B(3)$	0	$[5/8(2^{1/2})](n^3+3n^2+17n+15)B(1)$	$(5/16)(2n^4+4n^3+44n^2+44n+21)$
PQ^6P	$(-1/16)B(8)$	0	$(-1/8)(2n+7)B(6)$	0	$(-1/4)(n^2+5n)B(4)$	0	$(1/8)(2n^3+9n^2+79n+105)B(2)$	0	0

 m

$\hat{O}$	$n-1$	$n-2$	$n-3$	$n-4$	$n-5$	$n-6$	$n-7$	$n-8$
P^2	0	$(-1/2)A(1)$	0	0	0	0	0	0
PQP	$[1/2(2^{1/2})](A(0))^3$	0	$[-1/2(2^{1/2})]A(2)$	0	0	0	0	0
PQ^2P	0	0	0	$(-1/4)A(3)$	0	0	0	0
PQ^3P	$[1/4(2^{1/2})](2n^2+7)A(0)$	0	$[-1/4(2^{1/2})](n-1)A(2)$	0	$[-1/4(2^{1/2})]A(4)$	0	0	0
PQ^4P	0	$(1/8)(n^2-n+13)A(1)$	0	$(-1/8)(2n-3)A(3)$	0	$(-1/8)A(5)$	0	0
PQ^5P	$[5/8(2^{1/2})](n^3+14)A(0)$	0	$[-1/8(2^{1/2})](n^2-2n-18)A(2)$	0	$[-3/8(2^{1/2})](n-2)A(4)$	0	$[-1/8(2^{1/2})]A(6)$	0
PQ^6P	0	$(1/8)(2n^3-3n^2+67n-33)A(1)$	0	$(-1/4)(n^2-3n-4)A(3)$	0	$(-1/8)(2n-5)A(5)$	0	$(-1/16)A(7)$

^a Matrix elements are represented by $\langle n|\hat{O}|m\rangle$; $\hat{O}$ is the operator PQ^iP ; the $A(r)$ and $B(r)$ terms are defined in eq 52 and 53.

It should be pointed out that this procedure works well only if a good set of eigenvalues are known for the system and if the initially chosen set of force constants are close to the "true" constants. If the constants are not close to the "true" ones, the iterative procedure will most likely not converge to a minimum difference between observed and calculated frequencies.

Van Vleck Transformation Applied to Large-Amplitude Vibrations. The effective Hamiltonian for a large-amplitude vibration provides a means of describing a vibration independent of the other $3N - 7$ vibrations of the molecule. Given spectroscopic data for the large-amplitude mode in various quantum states of another vibration, the effective Hamiltonian may be derived in each of these states. Also, if a multidimensional Hamiltonian is known, the effective one-dimensional Hamiltonian(s) may be derived. This may be accomplished by the use of the Van Vleck transformation.^{4,18,36-43} The Van Vleck transformation is an approximate diagonalization technique which allows the factoring of an infinite-energy matrix into small submatrices which may be treated individually. The transformation is a perturbation technique and will, in this case, be taken through second order. An outline of the transformation to a specific two-dimensional problem involving one large-amplitude vibration and one small-amplitude vibration will be presented.

The Van Vleck transformation to second-order results in the matrix G with matrix elements given by^{16,42,43}

$$\langle m | G | m' \rangle = \langle m | H_0 | m' \rangle + \langle m | H_1 | m' \rangle + \frac{1}{2} \sum_n' \left[\frac{\langle m | H_1 | n \rangle \langle n | H_1 | m' \rangle}{E_m - E_n} + \frac{\langle m | H_1 | n \rangle \langle n | H_1 | m' \rangle}{E_{m'} - E_n} \right] \quad (61)$$

where the E 's are the energies of the designated states, H_0 is the zeroth-order Hamiltonian, H_1 is the perturbation Hamiltonian, and the prime on the summation indicates the sum is *not* over $n = m, m'$.

The total Hamiltonian for the large- and small-amplitude vibrations (neglecting pseudopotential terms) is given by

$$H = -\frac{\hbar^2}{2} \sum_r \sum_i \frac{\partial}{\partial q_r} g_{ri} \frac{\partial}{\partial q_i} - \frac{\hbar^2}{2} \sum_i \sum_j \frac{\partial}{\partial q_i} g_{ij} \frac{\partial}{\partial q_j} - \frac{\hbar^2}{2} \sum_r \sum_i \left[\frac{\partial}{\partial q_r} g_{ri} \frac{\partial}{\partial q_i} + \frac{\partial}{\partial q_i} g_{ri} \frac{\partial}{\partial q_r} \right] + V_L(q_L) + V_S(q_S) + \sum_{\text{cross terms}} f_L(q_L) f_S(q_S) \quad (62)$$

where M is the number of large-amplitude modes. The first term is the kinetic energy for the large-amplitude vibrations; the second term is the kinetic energy for small-amplitude vibrations; and the third term represents the interaction kinetic energy term between the large- and small-amplitude vibrations. The fourth and fifth terms are the potential energy for the large (L)- and small (S)-amplitude vibrations, respectively, and the sixth term is the cross term in the potential energy. The first, second, fourth, and fifth terms make up the H_0 Hamiltonian. The remaining constitute H_1 . The effect of a Van Vleck transformation on a problem involving a large- and small-amplitude vibration is depicted in Figure 4. Each block is represented by two quantum numbers for the small-amplitude vibration. The blocks are further subdivided and are labeled by two quantum numbers for the large-amplitude vibration.

Upon applying the Van Vleck transformation through second order (eq 61) to the total Hamiltonian in eq 62, we obtain the effective Hamiltonian for the large-amplitude vibrations given by the expression

$$H_{\text{eff}}^L = \left\{ -\frac{\hbar^2}{2} \sum_r \sum_i \left(\frac{\partial}{\partial q_r} g_{ri} \frac{\partial}{\partial q_i} \right) - \frac{\hbar^2}{2} \sum_r \sum_i \sum_j \left(\frac{\partial}{\partial q_r} g_{ri} + g_{ri} \frac{\partial}{\partial q_r} \right) \left\langle \phi_S^n \left| \frac{\partial}{\partial q_i} \right| \phi_S^{n'} \right\rangle - \frac{\hbar^2}{2} \sum_{n''} \frac{\left[\sum_r \sum_i \sum_j \left(\frac{\partial}{\partial q_r} g_{ri} + g_{ri} \frac{\partial}{\partial q_r} \right) \left\langle \phi_S^n \left| \frac{\partial}{\partial q_i} \right| \phi_S^{n''} \right\rangle \right]^2}{E_S^n - E_S^{n''}} \right\} + \left\{ V_L(q_L) + \sum_{\text{cross terms}} f_L(q_L) \langle \phi_S^n | f_S(q_S) | \phi_S^n \rangle + \frac{\left[\sum_{\text{cross terms}} f_L(q_L) \langle \phi_S^n | f_S(q_S) | \phi_S^n \rangle \right]^2}{\sum_{n''} \frac{E_S^n - E_S^{n''}}{E_S^n - E_S^{n''}}} \right\} + 2 \sum_{n''} \left\{ \frac{\left[\sum_{\text{cross terms}} f_L(q_L) \langle \phi_S^n | f_S(q_S) | \phi_S^{n''} \rangle \right]}{E_S^n - E_S^{n''}} \times \left[\frac{\hbar^2}{2} \sum_r \sum_i \sum_j \left(\frac{\partial}{\partial q_r} g_{ri} + g_{ri} \frac{\partial}{\partial q_r} \right) \left\langle \phi_S^n \left| \frac{\partial}{\partial q_i} \right| \phi_S^{n''} \right\rangle \right] \right\} + E_S \quad (63)$$

where we have assumed that $[\partial/\partial q_i, g_{ri}] = 0$ and noted that we are considering the $n = n'$ state (necessary for determining the effective Hamiltonian in a particular quantum state of the small-amplitude Hamiltonian). The ϕ_S^n 's are wave functions for the small-amplitude vibrations. These results agree with those presented elsewhere.¹⁸

We now consider the transformation as applied through "second order" to a two-dimensional vibrational problem with a potential function of the form given in eq 39. First, we transform to mass-weighted coordinates:

$$x = \mu_1^{-1/2} Q_1 \quad \text{and} \quad y = \mu_2^{-1/2} Q_2 \quad (64)$$

(Note we have redefined Q to be a mass-weighted rather than a dimensionless coordinate as given in eq 41.) This results in

$$V(Q_1, Q_2) = A_1 Q_1^4 + B_1 Q_1^2 + A_2 Q_2^4 + B_2 Q_2^2 + A_{13} Q_1 Q_2^3 + A_{22} Q_1^2 Q_2^2 + A_{31} Q_1^3 Q_2 \quad (65)$$

where

$$A_1 = \frac{a_1}{\mu_1^2}; \quad B_1 = \frac{b_1}{\mu_1}; \quad A_2 = \frac{a_2}{\mu_2^2}; \quad B_2 = \frac{b_2}{\mu_2}$$

$$A_{13} = \frac{a_{13}}{(\mu_1 \mu_2^3)^{1/2}}; \quad A_{22} = \frac{a_{22}}{(\mu_1 \mu_2)}; \quad A_{31} = \frac{a_{31}}{(\mu_1^3 \mu_2)^{1/2}} \quad (66)$$

and $a_1, b_1, a_2, b_2, a_{13}, a_{22}$, and a_{31} are those given in eq 39. If Q_2 is harmonic (and small amplitude) and we neglect the last cross term in eq 62, the effective one-dimensional potential function for the large-amplitude vibration becomes

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TABLE III: Harmonic Oscillator Matrix Elements for Q^r Operators^a

$\hat{O}$	m						
	$n+6$	$n+5$	$n+4$	$n+3$	$n+2$	$n+1$	n
Q	0	0	0	0	0	$(1/2^{1/2})B(1)$	0
Q^2	0	0	0	0	$(1/2)B(2)$	0	$(1/2)(2n+1)$
Q^3	0	0	0	$[1/2(2^{1/2})]B(3)$	0	$[3/2(2^{1/2})]B(1)^3$	0
Q^4	0	0	$(1/4)B(4)$	0	$(1/2)(2n+3)B(2)$	0	$(3/4)(2n^2+2n+1)$
Q^5	0	$[1/4(2^{1/2})]B(5)$	0	$[5/4(2^{1/2})](n+2)B(3)$	0	$[5/4(2^{1/2})](2n^2+4n+3)B(1)$	0
Q^6	$(1/8)B(6)$	0	$(3/8)(2n+5)B(4)$	0	$(15/8)(n^2+3n+3)B(2)$	0	$(5/8)(4n^3+6n^2+8n+3)$

$\hat{O}$	m					
	$n-1$	$n-2$	$n-3$	$n-4$	$n-5$	$n-6$
Q	$(1/2^{1/2})A(0)$	0	0	0	0	0
Q^2	0	$(1/2)A(1)$	0	0	0	0
Q^3	$[3/2(2^{1/2})]A(0)^3$	0	$[1/2(2^{1/2})]A(2)$	0	0	0
Q^4	0	$(1/2)(2n-1)A(1)$	0	$(1/4)A(3)$	0	0
Q^5	$[5/4(2^{1/2})](2n^2+1)A(0)$	0	$[5/4(2^{1/2})](n-1)A(2)$	0	$[1/4(2^{1/2})]A(4)$	0
Q^6	0	$(15/8)(n^2-n+1)A(1)$	0	$(3/8)(2n-3)A(3)$	0	$(1/8)A(5)$

^a Matrix elements are represented by $\langle n|\hat{O}|m\rangle$; $\hat{O}$ is the operator Q^r ; the $A(r)$ and $B(r)$ terms are defined in eq 52 and 53.

$$V_{\text{eff}}^L = \left(A_1 - \frac{A_{22}^2 \mu_2^2}{2\alpha^2 \hbar \nu_2} (m + \frac{1}{2}) - \frac{3A_{13}A_{31}\mu_2^2}{\alpha^2 \hbar \nu_2} (m + 1) \right) Q_1^4 + \left(B_1 + \frac{A_{22}\mu_2}{2\alpha} (2m + 1) - \frac{A_{13}2\mu_2^3}{8\alpha^3 \hbar \nu_2} (30m^2 + 30m + 11) \right) Q_1^2 - \frac{A_{31}^2 \mu_2^2}{2\alpha \hbar \nu_2} Q_1^6 \quad (67)$$

This expression may be simplified to

$$V_{\text{eff}}^L (\text{cm}^{-1}) = \left(A_1 - 586.464 \frac{A_{22}^2}{\bar{\nu}_2^3} (m + \frac{1}{2}) - 3410.78 \frac{A_{13}A_{31}}{\bar{\nu}_2^3} (m + 1) \right) Q_1^4 + \left(B_1 + 33.7330 \frac{A_{22}}{\bar{\nu}} (m + \frac{1}{2}) - 4789.68 \frac{A_{13}^2}{\bar{\nu}_2^4} (30m^2 + 30m + 11) \right) Q_1^2 - 16.8567 \frac{A_{31}^2}{\bar{\nu}_2^2} Q_1^6 \quad (68)$$

where Q_1^4 is in $\text{cm}^{-1} \text{u}^{-2} \text{\AA}^{-4}$, Q_1^2 in $\text{cm}^{-1} \text{u}^{-1} \text{\AA}^{-2}$, and Q_1^6 in $\text{cm}^{-1} \text{u}^{-3} \text{\AA}^{-6}$.

The kinetic energy part of the effective one-dimensional Hamiltonian, assuming the g_{45} terms are a function only of the large-amplitude coordinate, becomes

$$T_{\text{eff}}^L (\text{cm}^{-1}) = [-16.8567 g_{44}^{(0)} + 0.426984 \mu_2 (g_{45}^{(0)})^2] \mu_1 \frac{\partial^2}{\partial Q_1^2} - \frac{\hbar^2}{8\pi^2} \left\{ \sum_{i=1}^r \mu_1^{-1/2} g_{44}^{(i)} \frac{\partial}{\partial Q_1} Q_1^i \frac{\partial}{\partial Q_1} - \frac{\hbar^2 \mu_2}{16\pi^2} \sum_{i=1}^s \sum_{j=1}^w \mu_1^{-(i+j)/2} g_{45}^{(i+j)} \left[4 \frac{\partial}{\partial Q_1} Q_1^{(i+j)} \frac{\partial}{\partial Q_1} + 2(j-i) \frac{\partial}{\partial Q_1} Q_1^{(i+j-1)} + (2i^2 + ij - 2i) Q_1^{(i+j-2)} \right] \right\} \quad (69)$$

Here we have made use of the commutation relationship

$$\left[Q_1^i \frac{\partial}{\partial Q_1} \right] = -i Q_1^{i-1} \quad (70)$$

Although we have used the transformation for the case of one small- and one large-amplitude vibration, this can also be used for separation of two or more large-amplitude vibrations. The accuracy here would be dependent upon the degree of perturbation

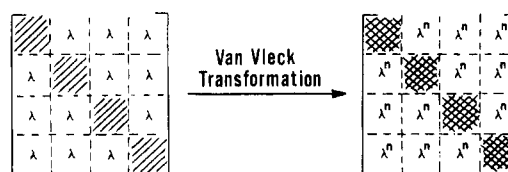


Figure 4. The effect of a Van Vleck transformation on a matrix H . The shaded areas are zeroth order in λ and the unshaded areas of first order or m th order. The blocks have different sets of quantum numbers for the small-amplitude vibration; each block is then subdivided into elements with different quantum numbers for the large-amplitude vibration.

one or several large-amplitude vibrations have on one another. The main utility of the Van Vleck transformation for solving two-dimensional potential energy surfaces is that an estimate can be obtained for the two-dimensional surface. If spectroscopic data are available for a particular vibration in two states (e.g. ground and first excited states of a second vibration), then the effective one-dimensional potential (and/or kinetic) functions can be calculated. Once the effective one-dimensional functions are known, Van Vleck theory can be used to estimate the two-dimensional function (particularly cross terms).

The determination of the potential energy surface for the ring-puckering (x) and PH inversion (y) vibrations of 3-phospholene⁹ represents an example of application of Van Vleck methods. In this particular case the effective one-dimensional ring puckering functions $V_g(x)$ and $V_e(x)$ were determined for the ground and first excited state of the PH inversion, respectively:

$$V_g(x) = a_g x^4 + b_g x^2 + c_g x^3 \quad (71)$$

and

$$V_e(x) = a_e x^4 + b_e x^2 + c_e x^3 \quad (72)$$

Use of the Van Vleck methods allowed the use of the effective potential constants in eq 71 and 72 to determine those for the two-dimensional function of the form

$$V(x,y) = a_1 x^4 + b_1 x^2 + c_1 x^3 + b_2 y^2 + a_{22} x^2 y^2 \quad (73)$$

Experimentally it was found that b_g and b_e were very similar in magnitude as were c_g and c_e . Consequently, the relationships $b_1 = b_g = b_e$ and $c_1 = c_g = c_e$ were used. In relation to eq 65, 66, and 68, it should also be noted that $A_2 = A_{13} = A_{31} = 0$ here and also that a_g corresponds to $m = 0$ and a_e to $m = 1$. Consequently, the coefficient of the Q_1^4 term may be evaluated to be

$$A_1 - 586.464 A_{22}^2 \bar{\nu}_2^{-3} (m + \frac{1}{2}) = \mu_1^{-2} [a_1 - 586.464 \mu_2^{-2} \bar{\nu}_2^{-3} a_{22}^2 (m + \frac{1}{2})] \quad (74)$$

The expression in brackets on the right-hand side represents the effective quartic coefficient (a_g or a_e depending on m) for eq 71 and 72 since $Q_1^4 = \mu_1^2 x^4$. From eq 74 using first $m = 0$ and then $m = 1$ and taking the difference and rearranging, we have

$$a_{22} = \frac{\mu_2 \bar{\nu}_2^{-3/2}}{23.8202} (a_g - a_e) \quad (75)$$

This provides a good approximation for a_{22} , which would otherwise be difficult to obtain.

Conclusion

We have presented a review of the development of the vibrational Hamiltonian from fundamental principles. We have also presented the formalism for quantum-mechanical calculations necessary to arrive at a potential energy surface representing the interaction between two vibrational modes, which can to a first approximation be treated separately from the other $3N - 8$ vibrations. The theoretical treatment, which is aimed at determining a specific potential energy surface, assumes the availability of

appropriate data from vibrational spectroscopy. The determination of a two-dimensional potential energy surface is necessary for a thorough understanding of the conformational dynamics of a molecule. The general procedures presented here have been widely used in our laboratory to determine kinetic and potential energy surfaces for a variety of small ring molecules.⁹⁻¹¹

Recently, the two-dimensional potential energy surface for 3-phospholene has been reported.⁹ The techniques presented here were used extensively to arrive at a potential energy surface for this molecule. Several approximations were made using the Van Vleck transformation, and these were useful in arriving at the final potential energy surface.

The techniques presented here are most useful for studying vibrations that are separable from all other vibrations in the molecule. The expansion of the techniques to the study of vibrations in systems other than small ring molecules has not been widely investigated, but this has considerable potential.

Acknowledgment. This work was sponsored by the National Science Foundation and the Robert A. Welch Foundation.

Clusters of Organic Molecules in a Supersonic Jet Expansion

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A wide variety of clusters were obtained in seeded supersonic conical jet expansions of organic molecules. It was shown that mixed clusters probably are formed statistically in the collisional region of the expansion, although kinetic effects cannot be excluded. The molecular fragmentation in the clusters upon mass selection using electron impact ionization is negligible, which makes the use of supersonic jet expansions under cluster forming conditions a valuable addition to standard mass spectrometry analytical techniques. The effect of the electron impact ionization on cluster fragmentation was studied.

Introduction

Clusters form aggregates which can be thought of as intermediate between the solid, the liquid, and the gas phase. A study of cluster properties may yield important information for such fields as the catalysis of chemical reactions and condensation and solvation effects, and clusters may even be used as a model for the study of surface phenomena. The properties of the clusters generated in a supersonic jet expansion can be studied by a range of optical, mass spectrometric, and electron excitation methods. In addition molecular dynamics calculations can provide us with a theoretical foundation for the experimental observations. By consequence the observation of cluster formation in a supersonic expansion¹ has opened up a whole new field of research, although only a limited amount of work in this area has been done so far.

Sievert et al.² produced small acetic acid clusters and analyzed them by electron impact (EI) mass spectrometry. They found extensive fragmentation upon ionization and a high yield of the $(M + H)^+$ peak which originates from the $(M - M)^+ \rightarrow (M + H)^+ + (M - H)$ cluster fragmentation reaction. Bernstein et al.³ studied toluene clusters in a supersonic beam and used EI and multiphoton ionization (MPI) mass spectrometry for detection and found less molecular and cluster fragmentation with the latter technique. An interesting observation was the different time

profiles in their pulsed experiments for the monomer and the higher clusters. Fung et al.⁴ generated mixtures of benzene- h_6 and benzene- d_6 clusters and analyzed the beam with MPI. They found that the excitation was mostly localized on one molecule of the cluster. Shinohara⁵ generated ammonia clusters and found more cluster formation at higher stagnation pressure and for a higher molecular weight of the carrier gas. Smalley et al.⁶ did some elegant experiments in which they generated metal atom clusters by laser evaporation, and their experiments resolved some old theoretical problems about the electronic properties of diatomics.

Finally, Stace⁷ recently showed extensive argon clustering around acetone and proposed a mechanism for their breakup upon electron ionization.

Other methods, such as secondary ion mass spectrometry⁸ and fast atom bombardment spectrometry⁹ have been used to generate

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